Hugo Armando Guillen Ramirez

Curriculum Vitae

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Current position

Postdoctoral researcher at the University of Bern, Bern, Switzerland.

Education

2014–2018 **PhD in Computer Science**, *Center for Scientific Research and Higher Education at Ensenada (CICESE Research Center)*, Baja California, Mexico.

2011–2014 **MSc in Computer Science**, *Center for Scientific Research and Higher Education at Ensenada (CICESE Research Center)*, Baja California, Mexico.

2007–2011 **Bachelor in Computer Systems Engineering**, *Instituto Tecnológico de Tepic*, Nayarit, Mexico.

Research interests

Genomic data science; long noncoding RNA; bioinformatics; machine learning; algorithm design; optimization; bio-inspired strategies; software development; graph theory.

Research experience

2019–present **Postdoctoral Researcher**, *Project: "Identification and functional prediction of lncRNA elements."*, Group leader: Dr. Rory Johnson, University of Bern, Department for BioMedical Research, Bern, Switzerland.

2019–2019 **Postdoctoral Researcher**, *Project: "Finding research opportunities in Artificial Intelligence for the public health of the Mexican population."*, Group leader: Dr. Salvador Villareal Reyes, CICESE Research Center/Artificial Intelligence Consortium, Mexico.

Doctoral thesis (2014–2018)

Title *Methods for non-coding RNA gene prediction* (translated from Spanish)

Description The main objective of this project is to develop a non-coding RNA (ncRNA) gene predictor based in homology search on known sequences supported by machine learning, meta-heuristics, and state-of-the-art biological information.

Supervisor Dr. Israel Marck Martínez Pérez, CICESE Research Center, Mexico

Master thesis (2011-2014)

Title *Design of a Tissue P-System and a molecular algorithm to solve the MAX-CLIQUE problem* (translated from Spanish)

Description This theoretical work presents a tissue P-System that solves the MAX-CLIQUE problem in polynomial time, and also a constructive DNA algorithm inspired by this system.

Supervisor Dr. Israel Marck Martínez Pérez, CICESE Research Center, Mexico

Publications

Articles in peer-reviewed journals

2021 Bosch-Guiteras, N., Uroda, T., **Guillen-Ramirez, H.A.**, Riedo, R., Gazdhar, A., Esposito, R., Pulido-Quetglas, C., Zimmer, Y., Medová, M. & Johnson, R. (2021). Enhancing CRISPR deletion via pharmacological delay of DNA-PKcs. Genome Research. https://doi.org/10.1101/gr.265736.120

2020 Aguilera-Mendoza, L., Marrero-Ponce, Y., García-Jacas, C. R., Chavez, E., Beltran, J. A., **Guillen-Ramirez, H. A.**, & Brizuela, C. A. (2020). Automatic construction of molecular similarity networks for visual graph mining in chemical space of bioactive peptides: an unsupervised learning approach. Scientific reports, 10(1), 1-23. https://doi.org/10.1038/s41598-020-75029-1

2019 Calvillo-Vázquez, J. G., **Guillén-Ramírez, H. A.**, DiazDuarte-Rodríguez, M., Licea-Claverie, A., & Méndez, E. R. (2019). Particle size distribution from extinction and absorption data of metallic nanoparticles. Applied optics, 58(36), 9955-9966. https://doi.org/10.1364/AO.58.009955

2019 Aguilera-Mendoza, L., Marrero-Ponce, Y., Beltran, J. A., Tellez Ibarra, R., **Guillen-Ramirez, H. A.**, & Brizuela, C. A. (2019). "Graph-based data integration from bioactive peptide databases of pharmaceutical interest: toward an organized collection enabling visual network analysis." Bioinformatics. https://doi.org/10.1093/bioinformatics/btz260

2018 **Guillen-Ramirez, H. A.**, & Martínez-Pérez, I. M. (2018). "Classification of riboswitch sequences using k-mer frequencies". Biosystems, 174, 63-76. https://doi.org/10.1016/j.biosystems.2018.09.001

Articles in proceedings

2018 Ibarra-Hernández, R. F., Bertin, N., Alonso-Arévalo, M. A., & **Guillén-Ramírez, H. A.** (2018). "A benchmark of heart sound classification systems based on sparse decompositions". In Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series (Vol. 10975). https://doi.org/10.1117/12.2506758

2017 **Guillen-Ramirez, H. A.**, Colbes, J., Brizuela, C. A., & Martinez-Perez, I. M. (2017). "Accurate classification of immunomodulatory RNA sequences". In 2017 International Joint Conference on Neural Networks (IJCNN) (pp. 236-241). IEEE. https://doi.org/10.1109/IJCNN.2017.7965860

2013 Brizuela, C. A., Luna-Taylor, J. E., Martinez-Perez, I., **Guillen, H. A.**, Rodriguez, D. O., & Beltran-Verdugo, A. (2013). "Improving an evolutionary multi-objective algorithm for the biclustering of gene expression data". In 2013 IEEE Congress on Evolutionary Computation (pp. 221-228). IEEE. https://doi.org/10.1109/CEC.2013.6557574

Fellowships, Grants & Awards

2021–2023 **Marie Skłodowska-Curie Individual Fellowship (H2020-MSCA-IF-2019)**, European Commission (€196,590.72). Project: RNADOMAIN – Computational genomics of long noncoding RNA domains across metazoans. Host institution: University College Dublin, Ireland.

2021–2024 "Candidate to National Researcher" member of the Mexican National System of Researchers (SNI) (Mexico).

2014–2018 CONACYT National Scholarship for doctoral studies (Mexico).

2011–2013 CONACYT National Scholarship for master studies (Mexico).

2011 Award for academic achievement (best grade in class 2006-2011). Instituto Tecnológico de Tepic (Mexico).

Invited talks

2020, Oct "Discrete mathematics and the design of CRISPR-Cas9 genome editing libraries" at Instituto Tecnológico de Tepic, Mexico.

2017, Mar "Score functions for discriminant k-mers and their application in ncRNA classification" (translated from Spanish) at CICESE Research Center, Mexico.

2016, Oct "Methods for ncRNA gene prediction: RNA secondary structure alternative graph representations" (translated from Spanish) in the XII National Meeting on Mathematical Biology (Center of Mathematical Sciences, UNAM Campus Morelia), Mexico.

2016, Oct "Bio-computing: RNA, BioPython, and Graphs" (translated from Spanish) in the Computer Science Fall School at CICESE Research Center, Mexico.

Conference presentations and posters

Poster **Guillen-Ramirez, H. A.** & Johnson, R. "Transposable elements as functional domains of long noncoding RNAs". Presented in the 2020 Virtual meeting on Transposable Elements (Cold Spring Harbor Laboratory).

Conference **Guillen-Ramirez, H. A.**, Colbes, J., Brizuela, C. A., & Martinez-Perez, I. M. (2017). "Accurate classification of immunomodulatory RNA sequences". International Joint Conference on Neural Networks 2017, Anchorage, Alaska, USA.

Poster **Guillen, H. A.** & Martinez-Perez, I. "Simulation of a programmable molecular computer". Presented in the 21st International Conference on DNA Computing and Molecular Programming 2015, Boston, USA.

Conference **Guillen, H. A.** & Martinez-Perez, I. "Challenges in the prediction of non-coding RNA genes" (translated from Spanish). In the proceedings of the National Meeting on Computer Science (Encuentro Nacional de Ciencias de la Computación, ENC 2014), Mexico.

Poster **Guillen, H. A.** & Martinez-Perez, I. "Design of a Tissue P-System for solving the MAX-CLIQUE problem" (translated from Spanish). Presented in the 6th International Congress on Computer Science (6to.Congreso Internacional en Ciencias Computacionales, CiComp 2013), Mexico.

Experience

Teaching

2019 **Computational Biology** course (Genome indexing algorithms) at CICESE Research Center, Mexico [Graduate level].

2017–2019 **Artificial Intelligence**, **Advanced Mathematics**, **Evolutionary Computing**, and **Assembly Language** courses at Universidad Autónoma de Baja California, Ensenada, Mexico [Undergraduate level].

2017–2018 **Mathematics II, IV, and V** (in English), **Intermediate Algebra I and II** (in English) and **Mathematical reasoning V and VI** (in Spanish) at Xochicalco High School, Ensenada, Mexico [High school level].

2012–2012 Teacher assistant of the **Discrete Mathematics** course at CICESE Research Center. [Graduate level].

Mentoring

2018–2019 Supervision of a team of undergraduate students at the Universidad Autónoma de Baja California (UABC), Mexico, to develop a research project on Artificial Intelligence. The results were presented by H. J. Chávez in a Mexican conference: *Barajas-Alcalá, R., Orozco-Lomelí, D. U., Sandoval-Chávez, H. J., & **Guillén-Ramírez, H. A.** (2019). Descubrimiento automático de reglas en juegos mediante Q-Learning (Automatic discovery of rules in games using Q-Learning (translated from Spanish.)). Revista Aristas: Investigación Básica y Aplicada, 7(14), 8.*

2018 Supervision of two teams (8 students in total) to develop projects for the "Engineering Day" at UABC: "An evolutionary algorithm for solving the videogame Rullo" and "An ant colony optimization algorithm (ACO) for solving the Traveling Salesman Problem".

2018 Instructor at the "Autumn School on Computer Science" at CICESE Research Center. Topics were related to RNA Bioinformatics. The audience were undergraduate students in their final year.

Developer

2019–present University of Bern, Bern, Switzerland. In my current postdoc project I have been developing tools for feature engineering, automatic machine learning, network visualization, statistical testing on genomic regions, and pipelines for designing and processing CRISPR deletion libraries.

2013–2018 CICESE Research Center, Ensenada, Mexico. During my doctoral work I developed and implemented machine learning techniques, meta-heuristics, and visualisation tools for classification problems involving genomic sequences and RNA secondary structures.

2016 Developer of REFRACTIVEINDEX.INFO-SQLITE, a Python 3 open source package to locally process the refractiveindex.info database. `http://github.com/HugoGuillen/refractiveindex.info-sqlite` (13 4 ★).

2012 MXGlobal Solutions, Nayarit, Mexico. I developed C# and SQL software.

2010-2011 Instituto Tecnológico de Tepic, Nayarit, Mexico. I developed a web system in Java/JSP.

Attendend courses

2021 Data Management Plan (Swiss Institute of Bioinformatics), streamed from Lausanne, Switzerland.

2020 Enrichment Analysis, (Swiss Institute of Bioinformatics), streamed from Lausanne, Switzerland.

2020 Advance statistics: statistical modeling, (Swiss Institute of Bioinformatics), Lausanne, Switzerland.

2016 XVIII Fall School on Mathematical Biology (Center of Mathematical Sciences, UNAM Campus Morelia), Michoacán, Mexico.

2016 NGS Data Analysis and Informatics Conference, San Diego, USA.

2015 Applications of Artificial Organic Networks (National Meeting on Computer Science 2015), Baja California, Mexico.

Service

Journal reviewer Molecular Genetics and Genomics (MGAG), Springer.

Technical skills

Programming languages Python, R, awk, bash scripting, SQL, C#, Java, Javascript, MATLAB, LaTeX

Databases SQLite, PostgreSQL, MySQL/MariaDB.

Containers Synergy container, Docker, conda environments.

Machine learning Weka, scikit-learn, R packages.

Graph theory NetworkX, Neo4j, Gephi, Graphviz

Bioinformatics Genome querying (bedtools, bedops, GIGGLE), alignment tools, RNA secondary structure prediction, RNA-Seq workflows, visualization (IGV, UCSC Genome Browser, pyGenomeTracks), CRISPR screen data analysis, feature engineering for genomic annotations.

Languages

Spanish: native speaker. **English:** proficient.